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Semiconductor structure and method for manufacturing semiconductor structure

Abstract

A semiconductor structure and a method for manufacturing a semiconductor structure are provided, which relate to the technical field of semiconductors. The semiconductor structure includes a substrate and a plurality of first conductive layers. The substrate includes a plurality of first trenches extending in a first direction and a plurality of second trenches extending in a second direction. A plurality of active pillars are provided between the plurality of first trenches and the plurality of second trenches. The first direction intersects with the second direction. Each of the plurality of first conductive layers is arranged on each of sidewalls, which are arrayed in the first direction, of a respective one of the plurality of active pillars.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This is a continuation application of International Patent Application No. PCT/CN2022/104041, filed on Jul. 6, 2022, which claims priority to Chinese Patent Application No. 202210018352.4, filed on Jan. 7, 2022 and entitled “SEMICONDUCTOR STRUCTURE AND METHOD FOR MANUFACTURING SEMICONDUCTOR STRUCTURE”. The disclosures of International Patent Application No. PCT/CN2022/104041 and Chinese Patent Application No. 202210018352.4 are incorporated by reference herein in their entireties.

BACKGROUND

(1) A Dynamic Random Access Memory (DRAM) is a semiconductor memory for writing and reading data randomly at high speed, which is widely used in data storage devices or apparatuses.

(2) The DRAM generally includes a plurality of repeated memory cells, and each memory cell includes a transistor and a capacitor. A source or a drain of the transistor is connected to the capacitor through a capacitor contact to write data information into the capacitor or read data information from the capacitor. However, the contact resistance between the source or the drain of the transistor and the capacitor contact is relatively large, which degrades the performance of the semiconductor structure.

SUMMARY

(3) The disclosure relates to the technical field of semiconductors, and in particular to a semiconductor structure and a method for manufacturing a semiconductor structure.

(4) Embodiments of the disclosure provide a semiconductor structure and a method for manufacturing a semiconductor structure, so as to reduce the contact resistance between a transistor and a capacitor contact, thereby improving the performance of the semiconductor structure.

(5) In order to achieve the purpose, the embodiments of the disclosure provide the following technical solutions.

(6) A first aspect of the embodiments of the disclosure provides a semiconductor structure, which includes: a substrate, in which the substrate includes a plurality of first trenches extending in a first

direction and a plurality of second trenches extending in a second direction, a plurality of active pillars are provided between the plurality of first trenches and the plurality of second trenches, and the first direction intersects with the second direction; and a plurality of first conductive layers, in which each of the plurality of first conductive layers is arranged on each of sidewalls, which are arrayed in the first direction, of a respective one of the plurality of active pillars.

(7) A second aspect of the embodiments of the disclosure provides a method for manufacturing a semiconductor structure, which includes the following operations.

(8) A substrate is provided.

(9) A plurality of first trenches extending in a first direction and a plurality of second trenches extending in a second direction are formed in the substrate, in which a plurality of active pillars are provided between the plurality of first trenches and the plurality of second trenches, and the first direction intersects with the second direction.

(10) A first conductive layer is formed on each of sidewalls, which are arrayed in the first direction, of each of the plurality of active pillars.

(11) In addition to the technical problems solved by the embodiments of the disclosure described above, the technical features constituting the technical solutions, and the beneficial effects brought by the technical features of these technical solutions, other technical problems to be solved by the semiconductor structure and the method for manufacturing the semiconductor structure provided in the embodiments of the disclosure, other technical features included in the technical solutions, and other beneficial effects brought by these technical features will be further described in detail in the detail description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) In order to describe the technical solutions in the embodiments of the disclosure or in the related art more clearly, the accompanying drawings required to be used in descriptions about the embodiments of the disclosure or the related art will be simply introduced below. Apparently, the accompanying drawings described below show merely some embodiments of the disclosure, and persons of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative effort.

(2) FIG. 1 is a top view of a semiconductor structure according to an embodiment of the disclosure;

(3) FIG. 2 is a cross-sectional view taken along the A-A' direction in FIG. 1;

(4) FIG. 3 is a cross-sectional view taken along the B-B' direction in FIG. 1;

(5) FIG. 4 is a flowchart of a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(6) FIG. 5 is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a first sub-trench in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(7) FIG. 6 is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a second sub-trench in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(8) FIG. 7 is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a third initial isolation layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(9) FIG. 8 is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a third isolation layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(10) FIG. 9 is a schematic cross-sectional view of a structure taken along the A-A' direction after

forming a first initial conductive layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(11) FIG. **10** is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a first conductive layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(12) FIG. **11** is a schematic cross-sectional view of a structure taken along the A-A' direction after removing a third isolation layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(13) FIG. **12** is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a first sub isolation layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(14) FIG. **13** is a schematic cross-sectional view of a structure taken along the A-A' direction after forming an insulation layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(15) FIG. **14** is a schematic cross-sectional view of a structure taken along the A-A' direction after forming a second sub isolation layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(16) FIG. **15** is a schematic cross-sectional view of a structure taken along the B-B' direction after forming a first trench in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(17) FIG. **16** is a schematic cross-sectional view of a structure taken along the B-B' direction after forming a first initial dielectric layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure;

(18) FIG. **17** is a schematic cross-sectional view of a structure taken along the B-B' direction after forming a first dielectric layer in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure; and

(19) FIG. **18** is a schematic cross-sectional view of a structure taken along the B-B' direction after forming a bit line structure in a method for manufacturing a semiconductor structure according to an embodiment of the disclosure.

REFERENCE NUMERALS

(20) **10**: Substrate; **11**: First trench; **12**: Second trench; **13**: Active pillar; **20**: First conductive layer; **21**: First initial conductive layer; **30**: First isolation layer; **31**: Void; **32**: First sub isolation layer; **33**: Second sub isolation layer; **34**: Insulation layer; **40**: Bit line structure; **50**: First dielectric layer; **51**: First initial dielectric layer; **60**: Second conductive layer; **70**: Second isolation layer; **80**: Third isolation layer; **81**: Third initial isolation layer; **82**: Third trench; **83**: Fourth trench.

DETAILED DESCRIPTION

(21) In the embodiments of the disclosure, the first conductive layer is provided on each of the sidewalls, which are arrayed in the first direction, of the respective one of the plurality of active pillars, so as to increase the size of the active pillar in the first direction, so that the purpose of increasing the area of the active pillar is achieved, and the contact resistance with the capacitor contact subsequently formed on the active pillar is reduced, thereby improving the performance of the semiconductor structure.

(22) In order to make the foregoing objectives, features and advantages of the embodiments of the disclosure more apparent and lucid, the technical solutions in the embodiments of the disclosure are clearly and completely described below with reference to the accompanying drawings in the embodiments of the disclosure. It is apparent that the embodiments described here are merely some rather than all of the embodiments of the disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the disclosure without creative efforts shall fall within the protection scope of the disclosure.

(23) In this embodiment, a semiconductor structure is not limited. The following description will be

made by taking the semiconductor structure as a DRAM as an example. However, this embodiment is not limited thereto, and the semiconductor structure in this embodiment may also be other structures. The semiconductor structure and a method for manufacturing a semiconductor structure will be described in detail below with reference to FIG. 1 to FIG. 18.

(24) As shown in FIG. 1 to FIG. 3, the semiconductor structure provided by the embodiments of the disclosure includes a substrate **10** and a plurality of first conductive layers **20**.

(25) The substrate **10** may be made of a semiconductor material. The semiconductor material may be one or more of silicon, germanium, a silicon germanium compound, or a silicon carbon compound.

(26) As shown in FIG. 1, the substrate **10** includes a plurality of first trenches **11** extending in a first direction and a plurality of second trenches **12** extending in a second direction. A plurality of active pillars **13** are provided between the plurality of first trenches **11** and the plurality of second trenches **12**. The first direction intersects with the second direction.

(27) It should be noted that the first direction is the Y direction shown in FIG. 1, and the second direction is the X direction shown in FIG. 1.

(28) The plurality of first trenches **11** are spaced apart from each other in the second direction, and the plurality of second trenches **12** are spaced apart from each other in the first direction, so as to divide the substrate **10** into the plurality of active pillars **13** spaced apart from each other. In this embodiment, the first trench **11** can be understood as an area between two lateral dotted lines as shown in FIG. 1, and the second trench **12** can be understood as an area between two longitudinal dotted lines as shown in FIG. 1.

(29) A depth of the first trench **11** may be the same as or different from a depth of the second trench **12**. Exemplarily, the depth of the first trench **11** is less than the depth of the second trench **12**.

(30) With reference to FIG. 1 and FIG. 2, each of the plurality of first conductive layers **20** is arranged on each of sidewalls, which are arrayed in the first direction, of a respective one of the plurality of active pillars **13**. That is, each first conductive layer **20** is arranged on each of sidewalls, which face the second trenches **12**, of a respective one of the plurality of active pillars **13**. Taking the orientation shown in FIG. 1 as an example, the first conductive layers **20** are respectively arranged on a front sidewall and a rear sidewall of each active pillar **13**. A material of the first conductive layer **20** includes polycrystalline silicon.

(31) In this embodiment, the first conductive layer **20** is arranged on each of the sidewalls, which face the second trenches **12**, of a respective one of the plurality of active pillars **13**, so as to increase the width of the active pillar in the first direction, so that the cross-sectional area of the top portion of the active pillar **13** can be increased. In this way, the contact area with the capacitor contact subsequently formed on top surface of the active pillar **13** is reduced, so that the contact resistance between the active pillar **13** and the capacitor contact is reduced, thereby improving the performance of the semiconductor structure.

(32) In addition, the first conductive layer **20** is arranged on each of sidewalls, which are arrayed in the first direction, of a respective one of the plurality of first active pillars **13**. That is, there is a gap between the bottom surface of the first conductive layer **20** facing the substrate **10** and the bottom surface of the second trench **12**, so that a subsequently formed bit line structure can be prevented from being electrically connected to the top portion of the active pillar **13**, thereby improving the yield of the semiconductor structure.

(33) In some embodiments, with reference to FIG. 2, each of the plurality of second trenches **12** includes a first isolation layer **30** and a void **31**. That is, the first isolation layer **30** is arranged in the second trench **12**, and the void **31** is arranged in the first isolation layer **30**, so that the dielectric constant of the first isolation layer **30** can be reduced, and the parasitic capacitance between adjacent bit line structures **40** can be further reduced, thereby improving the performance of the semiconductor structure.

(34) In this embodiment, the shape of the void **31** may be regular or irregular, which is influenced

by the shape of the second trench **12** and the process of forming the first isolation layer **30**. In addition, there may be one or more voids **31** in the same second trench **12**, which is not specifically limited in this embodiment.

(35) Exemplarily, taking the orientation shown in FIG. 2 as an example, and taking a section perpendicular to the substrate **10** as a longitudinal section, the shape of the longitudinal section of the second sub-trench **122** may be a droplet shape, which may facilitate the formation of the void **31**, thereby reducing the parasitic capacitance between the adjacent bit line structures **40**, while avoiding the electrical connection between the adjacent bit line structures **40**.

(36) In addition, the top surface of the first isolation layer **30** is lower than the top surface of the active pillar **13**, so that a space is formed between the top surface of the first isolation layer **30** and the top surface of the active pillar **13**, which facilitates the formation of a word line structure in this space.

(37) In some embodiments, the void **31** may be arranged below the first conductive layer **20**, or may be arranged between the first conductive layers **20**. In a case that the void **31** is arranged below the first conductive layer **20**, it can be understood that the void **31** is formed in a portion of the first isolation layer **30** between the bottom portion of the first conductive layer **20** and the bottom portion of the second trench **12**, and at least a portion of the void **31** is arranged between the adjacent bit line structures **40**, so as to ensure that the parasitic capacitance between the adjacent bit line structures **40** can be reduced.

(38) Correspondingly, in a case that the void **31** is arranged between the first conductive layers **20**, it can be understood that the void **31** is formed in a portion of the first isolation layer **30** between the bottom portion of the first conductive layer **20** and the top portion of the first conductive layer **20**, and at least a portion of the void **31** is arranged between the adjacent bit line structures **40**, so as to ensure that the parasitic capacitance between the adjacent bit line structures **40** can be reduced.

(39) In some embodiments, with reference to FIG. 2, each of the plurality of second trenches **12** includes a first sub-trench **121** and a second sub-trench **122**. The first sub-trench **121** is arranged above the second sub-trench **122**, and communicates with the second sub-trench **122**.

(40) Taking the orientation shown in FIG. 2 as an example, from top to bottom, the area arranged in the second trench **12** between the first dotted line and the second dotted line is the first sub-trench **121**, and the area arranged in the second trench **12** between the second dotted line and the third dotted line is the second sub-trench **122**.

(41) A cross-sectional area of the first sub-trench **121** is greater than or equal to a cross-sectional area of the second sub-trench **122**. In this embodiment, the cross-sectional area refers to the cross-sectional area in the direction parallel to the substrate **10**.

(42) In this embodiment, the second trench **12** has a structure with a large top portion and a small bottom portion. In this way, when the first isolation layer **30** is subsequently formed in the second trench **12**, under the influence of the cross-sectional areas of the first sub-trench **121** and the second sub-trench **122**, the smaller the cross-sectional area is, the more easily the void **31** is formed in the first isolation layer **30**.

(43) Taking the section perpendicular to the substrate **10** as the longitudinal section, the shape of the longitudinal section of the first sub-trench **121** may be rectangular, and the shape of the longitudinal section of the second sub-trench **122** can be regular or irregular. Exemplarily, a width of the second sub-trench **122** is gradually reduced from the end close to the first sub-trench **121** to the end away from the first sub-trench **121**. Taking the orientation shown in FIG. 2 as an example, the width of the second sub-trench **122** is gradually reduced from top to bottom, so that the second sub-trench **122** has a structure with a large top portion and a small bottom portion. For example, the shape of the longitudinal section of the second sub-trench **122** is V shape, so that the closer to the bottom portion of the substrate **10** is, the less the width of the second sub-trench **122** is, and the more easily the void **31** is formed in the second sub-trench **122**, thereby reducing the dielectric constant of the first isolation layer **30** arranged in the second sub-trench **122**.

(44) It should be noted that the shape of the longitudinal section of the second sub-trench **122** is not limited to the above-mentioned V shape, and the section of the second sub-trench **122** may also be a trapezoidal structure with a large top portion and a small bottom portion, so that a void **31** may also be formed in the second trench **12**.

(45) In some embodiments, each of the plurality of first conductive layers **20** is arranged on a sidewall, which is arranged in the first sub-trench **121**, of a respective one of the plurality of active pillars **13**. In a direction perpendicular to the substrate **10**, the active pillar **13** includes a channel area, a source area arranged on one side of the channel area, and a drain area arranged on the other side of the channel area. The drain area is arranged at the bottom portion of the active pillar **13** and is configured for being connected to the bit line structure. The channel area is arranged in the middle portion of the active pillar **13** and is configured for being connected to the word line. The source area is arranged on the top portion of the active pillar **13** and is configured for being connected to the capacitor contact, so as to achieve the electrical connection between the capacitor and the source area.

(46) Based on this, in this embodiment, the first conductive layer **20** is arranged on the sidewall, which is arranged in the first sub-trench **121**, of the active pillar **13**. In this way, on the one hand, the area of the source area of the active pillar **13** can be increased, and on the other hand, the manufacturing cost of the first conductive layer **20** can be reduced compared with the technical solution in which all the first conductive layers **20** are arranged on the sidewalls, which are arranged in the second trenches **12**, of the active pillars **13**.

(47) In some embodiments, with reference to FIG. 2 and FIG. 3, a bit line structure **40** is provided below a bottom portion of each of the plurality of first trenches **11**. The bit line structure **40** extends in the first direction.

(48) The bit line structure **40** is configured for being connected to the drain of the active pillar **13**, so that the data in the capacitor is read or the data is written into the capacitor through the bit line structure **40**. A material of the bit line structure **40** can include cobalt silicide or silicon germanium.

(49) In some embodiments, as shown in FIG. 3, a first dielectric layer **50** and a second conductive layer **60** are provided on a sidewall, which is arranged in a respective one of the plurality of first trenches **11**, of each of the plurality of active pillars **13**. The first dielectric layer **50** and the second conductive layer **60** are stacked on one another on the sidewall, which is arranged in the first trench **11**, of the active pillar **13**, and the first dielectric layer **50** is connected to the sidewall, which is arranged in the first trench **11**, of the active pillar **13**. The first dielectric layer **50** is configured as a gate oxide layer of a transistor subsequently formed, and a material of the first dielectric layer can be silicon oxide or other materials with high dielectric constant.

(50) In the section taken along the second direction, the second conductive layer **60** is arranged on the first dielectric layer **50**, and the projection of the second conductive layer **60** on the active pillar **13** at least covers the channel area of the active pillar **13**. In the section taken along the first direction, the second conductive layer **60** is arranged above the first isolation layer **30** of the second trench **12**, and the top surface of the second conductive layer **60** is lower than the top surface of the active pillar **13**, so that the second conductive layer **60** forms a word line extending in the second direction. The word line is configured for connecting all the active pillars arrayed in the same second direction, so that a voltage is applied to the channel area of the active pillar **13** through the word line to control connection or disconnection between the source area and the drain area of the active pillar **13**. A material of the second conductive layer **60** may include metal tungsten.

(51) In some embodiments, the semiconductor structure includes a second isolation layer **70**. As shown in FIG. 2, in the section taken along the first direction, the second isolation layer **70** is arranged above the second conductive layer **60** and arranged in a respective one of the plurality of second trenches **12**. As shown in FIG. 3, in the section taken along the second direction, the second isolation layer **70** is arranged between sidewalls, which are arranged in a respective one of the plurality of first trenches **11**, of two adjacent active pillars **13** of the plurality of active pillars. In

this way, any two adjacent second conductive layers **60** can be insulated from each other. The second isolation layer **70** can include an insulating material, such as silicon oxide or silicon nitride. (52) As shown in FIG. 4, an embodiment of the disclosure further provides a method for manufacturing a semiconductor structure, which includes the following operations.

(53) In **S100**, a substrate is provided.

(54) The substrate **10** may be made of a semiconductor material. The semiconductor material may be one or more of silicon, germanium, a silicon germanium compound, or a silicon carbon compound.

(55) In **S200**, a plurality of first trenches extending in a first direction and a plurality of second trenches extending in a second direction are formed in the substrate, in which a plurality of active pillars are provided between the plurality of first trenches and the plurality of second trenches, and the first direction intersects with the second direction intersect. The structure formed in this operation is shown in FIG. 5 and FIG. 15.

(56) It should be noted that in this embodiment, the forming sequence of the first trenches **11** and the second trenches **12** is not limited. The process of forming the first trenches **11** and the second trenches **12** may be performed in such a way that one of the first trenches **11** and the second trenches **12** is firstly etched, and then the other one of the first trenches **11** and the second trenches **12** is etched; or may also be any other process capable of forming the first trenches **11** and the second trenches **12**.

(57) For ease of description, in this embodiment, the process of manufacturing the semiconductor structure is described by taking the process in which the second trenches **12** are firstly formed as an example.

(58) Exemplarily, a first photoresist layer can be formed on the substrate **10** through a coating process, and then the first photoresist layer is patterned through an exposure, development or etching process to form a first mask pattern in the first photoresist layer. The first mask pattern includes a plurality of first mask protrusions and first mask openings arranged between any two adjacent first mask protrusions. The plurality of first mask protrusions are spaced apart from each other in the first direction, and each first mask protrusion extends in the second direction. The first direction is perpendicular to the second direction.

(59) As shown in FIG. 5, after the first mask openings are formed, a portion of a thickness of the substrate **10** exposed in the first mask openings may be removed by using an etching liquid or an etching gas, so as to form a plurality of first sub-trenches **121** in the substrate **10**. The plurality of first sub-trenches **121** are spaced apart from each other in the first direction, and each first sub-trench **121** extends in the second direction.

(60) As shown in FIG. 6, after the first sub-trenches **121** are formed, the first sub-trenches **121** are continuously etched to form the second sub-trenches **122** at the bottom portions of the first sub-trenches **121**. That is, a portion of the substrate **10** exposed at the bottom portions of the first sub-trenches **121** is removed in the vertical direction to form the second sub-trenches **122**. The second sub-trench **122** communicates with the first sub-trench **121**, so that the second trenches **12** are formed. The first sub-trench **121** is the area between the dotted line in FIG. 6 and the top portion of the second trench **12**, and the second sub-trench **122** is the area between the dotted line in FIG. 6 and the bottom portion of the second trench **12**.

(61) The cross-sectional area of the first sub-trench **121** is greater than or equal to the cross-sectional area of the second sub-trench **122**, so that the second trench **12** has a structure with a large top portion and a small bottom portion. In this way, when the first isolation layer **30** is subsequently formed in the second trench **12**, under the influence of the cross-sectional areas of the first sub-trench **121** and the second sub-trench **122**, the smaller the cross-sectional area is, the more easily the void **31** is formed in the first isolation layer **30**.

(62) In **S300**, a first conductive layer is formed on each of sidewalls, which are arrayed in the first direction, of each of the plurality of active pillars.

(63) As shown in FIG. 7, in the section taken along the first direction, a third initial isolation layer **81** is formed in the plurality of second trenches **12**. For example, the third initial isolation layer **81** may be formed in the second trenches **12** through an atomic layer deposition process. The third initial isolation layer **81** defines a third trench **82** in each second trench **12**, and the third initial isolation layer **81** also covers the top surface of the substrate **10**.

(64) As shown in FIG. 8, the third trench **82** may be etched to remove the third initial isolation layer **81** arranged on the top surface of the substrate **10** and in a portion of each second trench **12**, so that the remaining portion of the third initial isolation layer **81** forms a third isolation layer **80** arranged at the bottom portion of each second trench **12**. The third isolation layer **80** and the second trench define a fourth trench **83**.

(65) In this embodiment, a material of the third isolation layer **80** includes silicon oxide. The third isolation layer **80** is formed at the bottom portion of each second trench **12**, so that it can be ensured that the first conductive layer **20** is only formed on the upper portion of each second trench **12**, that is, it can be ensured that the first conductive layer **20** covers at least a portion of the sidewall of each first sub-trench **121**.

(66) As shown in FIG. 9, a first initial conductive layer **21** is formed in the fourth trenches **83** and on the substrate **10** through a deposition process. That is, the first initial conductive layer **21** is formed on a portion of the sidewall of each second trench **12** and on the top surface of the third isolation layer **80**. The first initial conductive layer **21** extends out of the second trenches **12** and covers the top surface of the substrate **10**.

(67) The deposition process may include any one of a Chemical Vapor Deposition (CVD) process, a Low Pressure CVD (LPCVD) process, a Plasma Enhanced CVD (PECVD) process, an Atomic Layer Deposition (ALD) process, or a Plasma Enhanced ALD (PEALD) process.

(68) As shown in FIG. 10, the first initial conductive layer **21** on the top surface of the third isolation layer **80** and the first initial conductive layer **21** on the top surface of the substrate **10** are removed by using an etching gas or an etching liquid, so that the first initial conductive layer **21** arranged on a sidewall, which is arranged in a respective one of the plurality of second trenches **12**, of each of the plurality of active pillars **13** is remained to form the first conductive layer **20**. In other words, the first initial conductive layer **21** arranged on the sidewall of each fourth trench **83** is remained to form the first conductive layer **20**.

(69) In this embodiment, a portion of the first initial conductive layer **21** arranged on an inner wall of each first sub-trench **121** may be remained, or all of the first initial conductive layer **21** arranged on an inner wall of each first sub-trench **121** may be remained. A material of first conductive layer **20** includes polycrystalline silicon.

(70) After the first conductive layer **20** is formed, as shown in FIG. 11, the third isolation layer **80** is removed.

(71) In this embodiment, the first conductive layer **20** is formed on a sidewall, which is arranged in a respective one of the plurality of second trenches **12**, of each of the plurality of active pillars **13**, so that the cross-sectional area on the top portion of the active pillar **13** can be increased, the contact area between the active pillar and the capacitor contact can be increased, and the contact resistance between the active pillar and the capacitor contact can be reduced, thereby improving the performance of the semiconductor structure.

(72) In some embodiments, after the first conductive layer is formed, a first isolation layer **30** and a void **31** are formed in each of the plurality of second trenches **12**. The first isolation layer **30** includes a first sub isolation layer **32** and a second sub isolation layer **33**.

(73) Exemplarily, the first isolation layer **30** is manufactured in two operations. As shown in FIG. 12, the first sub isolation layer **32** is firstly formed in the second sub-trench **122** through a thermal oxidation process, in which the second sub-trench **122** is not completely filled with the first sub isolation layer **32**.

(74) Next, as shown in FIG. 13, an insulation material with poor filling capability is deposited in

the first sub-trenches **121** to form an insulation layer **34**. The insulation layer **34** extends out of the first sub-trenches **121** and covers the top surface of the substrate **10**. In this embodiment, the area defined by the first sub isolation layer **32** and the first conductive layer **20** is not completely filled with the insulation layer **34**, so that the insulation layer **34** and the first sub isolation layer **32** define a void **31** in the second trench **12**.

(75) Next, as shown in FIG. **14**, a portion of a thickness of the insulation layer **34** is removed to expose a portion of the second trench **12**, so that a remaining portion of the insulation layer **34** forms the second sub isolation layer **33**. The second sub isolation layer **33** is connected to the first sub isolation layer **32** to form the first isolation layer **30**.

(76) In this embodiment, the void **31** is formed in the second trench **12**. In this way, on the one hand, the dielectric constant of the first isolation layer **30** can be reduced, on the other hand, the insulating property of the first isolation layer **30** can be increased, so that when the bit line structures are subsequently formed in the substrate, the parasitic capacitance between the adjacent bit line structures can be reduced, while avoiding the electrical connection between the adjacent bit line structures.

(77) In some embodiments, after the first isolation layer **30** is formed, the substrate **10** may be patterned again to form a plurality of first trenches **11** in the substrate **10**. The structures of the plurality of first trenches are shown in FIG. **15**. A depth of the first trench **11** is less than a depth of the second trench **12**, and the substrate is divided into a plurality of initial active pillars by the plurality of first trenches **11** and the plurality of second trenches **12**. And then each initial active pillar is ion doped through an ion implantation process to form the active pillar **13**.

(78) Exemplarily, firstly, the ion implantation energy and the type of implanted doping ions in the ion implantation process can be controlled to form a drain area at the bottom portion of the initial active pillar. Then, the ion implantation energy and the type of implanted doping ions in the ion implantation process can be controlled to form a channel area in the middle portion of the initial active pillar. Finally, the ion implantation energy and the type of implanted doping ions in the ion implantation process can be controlled to form a drain area on the top portion of the initial active pillar. The type of the doping ions in the drain area can be the same as the type of the doping ions in the source area. For example, the doping ions can include N-type ions. The type of the doping ions in the channel area is different from the type of the doping ions in the drain area. For example, the doping ions may include P-type ions.

(79) It should be noted that the process of forming the first trench **11** may be the same as the forming process of the second trench **12**, which is not repeated in this embodiment.

(80) As shown in FIG. **16**, after the first trenches **11** are formed, a first initial dielectric layer **51** may be deposited in each of the plurality of first trenches **11** through a deposition process. The first initial dielectric layer **51** is formed on the inner walls of the first trenches **11** and extends out of the first trenches **11** and covers the top surfaces of the active pillars **13**.

(81) The deposition process includes, but is not limited to, an ALD process and a PEALD process.

(82) As shown in FIG. **17**, the first initial dielectric layer **51** on the bottom portion of each first trench **11** and the first initial dielectric layer **51** on the top surface of each active pillar **13** are removed, so that the first initial dielectric layer **51** on the sidewall of each first trench **11** is remained to form the first dielectric layer **50**.

(83) In this operation, the active pillars **13** can be protected through the arrangement of the first dielectric layer **50**, so that the active pillars are prevented from being damaged in the ion implantation process, thereby improving the performance of the semiconductor structure.

(84) Next, as shown in FIG. **18**, the bottom portion of each first trench **11** is doped to form a bit line structure **40** below the bottom portion of each first trench **11**. The bit line structure **40** extends in the first direction. Each bit line structure **40** is connected to a different active pillar. The width of the bit line structure can be set by those skilled in the art, so that the bit line structure can connect two adjacent active pillars as much as possible and as evenly as possible, and the bit line structures

are not connected to each other.

(85) As an optical implementation of the bit line structure **40**, metal germanium may be deposited into the first trench **11**. Under certain reaction conditions, germanium reacts with silicon at the bottom portion of the active pillar **13** to form silicon germanium, so that the bottom portion of each active pillar **13** in the same second direction may form one bit line structure **40**.

(86) As an optical implementation of the bit line structure **40**, metal cobalt may be deposited into the bottom portion of the first trench **11**. Under certain reaction conditions, metal cobalt reacts with silicon at the bottom portion of the active pillar **13** to form cobalt silicide, so that the bottom portion of each active pillar **13** in the same second direction may form one bit line structure **40**.

(87) In some embodiments, with reference to FIG. 2 and FIG. 3, after the bit line structure **40** is formed, a second conductive layer **60** is formed on a side wall, which is arranged in a respective one of the plurality of first trenches, of each of the plurality of active pillars **13**. In the section taken along the first direction, the second conductive layer **60** is arranged above the first isolation layer **30** and arranged in a respective one of the plurality of second trenches **12**. In the section taken along the second direction, the second conductive layer **60** is arranged on a side of the first dielectric layer **50** away from the active pillar **13**.

(88) In this way, the second conductive layer **60** forms a word line extending in the second direction. The word line is configured for connecting all the active pillars arrayed in the second direction, so that a voltage is applied to the channel area of the active pillar **13** through the word line to control connection or disconnection between the source area and the drain area of the active pillar **13**. A material of the second conductive layer **60** may include metal tungsten.

(89) It should be noted that the process of forming the second conductive layer **60** in this embodiment may have various options. For example, a second initial conductive layer completely filling the first trench **11** and the second trench **12** is formed in the first trench **11** and the second trench **12**, and then removing a portion of the second initial conductive layer in the first trench **11** along the second direction to remain the second initial conductive layer on the inner wall of the first trench to form the second conductive layer **60**.

(90) In some embodiments, with reference to FIG. 2 and FIG. 3, in the section taken along the first direction, a second isolation layer **70** is formed above the second conductive layer **60** in a respective one of the second trenches. In the section taken along the second direction, the second isolation layer **70** is formed between the sidewalls, which are arranged in a respective one of the plurality of first trenches **11**, of two adjacent active pillars **13**.

(91) In this operation, the second isolation layer **70** may be formed in the first trench **11** and the second trench **12** through a deposition process, so that any two adjacent second conductive layers **60** can be insulated from each other through the second isolation layer **70**.

(92) Respective embodiments or implementation in the description are described in a progressive manner, each embodiment focuses on the difference from other embodiments, and the same and similar parts among the various embodiments are referenced to each other.

(93) In the description of the specification, the descriptions made with reference to terms “an implementation”, “some implementations”, “exemplary implementation”, “example”, “specific example”, “some examples” or the like refer to specific features, structures, materials or characteristics described in combination with the implementations or the examples are included in at least one implementation or example of the disclosure.

(94) In the specification, the schematic representations of the above terms do not necessarily refer to the same implementation or example. Moreover, the specific features, structures, materials or characteristics described herein may be combined with each other in a proper manner in any one or more implementations or examples.

(95) Finally, it should be noted that the above embodiments are merely used for describing rather than limiting the technical solutions of the present disclosure. Although the present disclosure is described in detail with reference to the foregoing embodiments, those of ordinary skill in the art

should understand that the technical solutions specified in the above embodiments can still be modified, or some or all of the technical features therein can be equivalently substituted; and such modifications or substitutions do not make the essence of the corresponding technical solutions depart from the scope of the technical solutions of each of the embodiments of the present disclosure.

Claims

1. A semiconductor structure, comprising: a substrate, wherein the substrate includes a plurality of first trenches extending in a first direction and a plurality of second trenches extending in a second direction, a plurality of active pillars are provided between the plurality of first trenches and the plurality of second trenches, and the first direction intersects with the second direction; and a plurality of first conductive layers, wherein each of the plurality of first conductive layers is arranged on each of sidewalls, which are arrayed in the first direction, of a respective one of the plurality of active pillars, the first conductive layer is arranged on a top surface of the active pillars, and there is a gap between a bottom surface of the first conductive layer facing the substrate and a bottom surface of the second trench.
2. The semiconductor structure according to claim 1, wherein each of the plurality of second trenches comprises a first isolation layer and a void.
3. The semiconductor structure according to claim 2, wherein the void is arranged below a respective one of the plurality of first conductive layers.
4. The semiconductor structure according to claim 2, wherein the void is arranged between two respective first conductive layers of the plurality of first conductive layers.
5. The semiconductor structure according to claim 2, wherein each of the plurality of second trenches comprises a first sub-trench and a second sub-trench, and the first sub-trench is arranged above the second sub-trench, and communicates with the second sub-trench.
6. The semiconductor structure according to claim 5, wherein a cross-sectional area of the first sub-trench is greater than or equal to a cross-sectional area of the second sub-trench.
7. The semiconductor structure according to claim 5, wherein each of the plurality of first conductive layers is arranged on a sidewall, which is arranged in the first sub-trench, of a respective one of the plurality of active pillars.
8. The semiconductor structure according to claim 2, wherein a bit line structure is provided below a bottom portion of each of the plurality of first trenches, wherein the bit line structure extends in the first direction.
9. The semiconductor structure according to claim 8, wherein a first dielectric layer and a second conductive layer are provided on a sidewall, which is arranged in a respective one of the plurality of first trenches, of each of the plurality of active pillars.
10. The semiconductor structure according to claim 9, wherein in a section taken along the first direction, the second conductive layer is arranged above the first isolation layer and arranged in a respective one of the plurality of second trenches.
11. The semiconductor structure according to claim 10, further comprising a second isolation layer, wherein in the section taken along the first direction, the second isolation layer is arranged above the second conductive layer and arranged in a respective one of the plurality of second trenches; and in a section taken along the second direction, the second isolation layer is arranged between sidewalls, which are arranged in a respective one of the plurality of first trenches, of two adjacent active pillars of the plurality of active pillars.
12. The semiconductor structure according to claim 1, wherein a depth of each of the plurality of first trenches is less than a depth of each of the plurality of second trenches.
13. A method for manufacturing a semiconductor structure, comprising: providing a substrate; forming, in the substrate, a plurality of first trenches extending in a first direction and a plurality of

second trenches extending in a second direction, wherein a plurality of active pillars are provided between the plurality of first trenches and the plurality of second trenches, and the first direction intersect with the second direction; and forming a first conductive layer on each of sidewalls, which are arrayed in the first direction, of each of the plurality of active pillars, the first conductive layer is arranged on a top surface of the active pillars, and there is a gap between a bottom surface of the first conductive layer facing the substrate and a bottom surface of the second trench.

14. The method for manufacturing the semiconductor structure according to claim 13, wherein in a section taken along the first direction, after a third isolation layer is formed at a bottom portion of each of the plurality of second trenches, the method further comprises: depositing a first initial conductive layer in each of the plurality of second trenches and on the third isolation layer, and removing a portion of the first initial conductive layer to remain the first initial conductive layer arranged on a sidewall, which is arranged in a respective one of the plurality of second trenches, of each of the plurality of active pillars, to form the first conductive layer.

15. The method for manufacturing the semiconductor structure according to claim 14, wherein each of the plurality of second trenches comprises a first sub-trench and a second sub-trench, and wherein the method further comprising: forming, in the substrate, a plurality of first sub-trenches extending in the second direction, and forming the second sub-trench at a bottom portion of a respective one of the plurality of first sub-trenches, wherein the second sub-trench communicates with the respective one of the plurality of first sub-trenches.

16. The method for manufacturing the semiconductor structure according to claim 15, wherein a cross-sectional area of the first sub-trench is greater than or equal to a cross-sectional area of the second sub-trench, and wherein the method further comprises: forming a first isolation layer and a void in each of the plurality of second trenches.

17. The method for manufacturing the semiconductor structure according to claim 16, further comprising: forming a second conductive layer on a sidewall, which is arranged in a respective one of the plurality of first trenches, of each of the plurality of active pillars, wherein in a section taken along the first direction, the second conductive layer is arranged above the first isolation layer and arranged in a respective one of the plurality of second trenches.

18. The method for manufacturing the semiconductor structure according to claim 17, further comprising: in the section taken along the first direction, forming, in each of the plurality of second trenches, a second isolation layer above the second conductive layer; and in a section taken along the second direction, forming the second isolation layer between sidewalls, which are arranged in a respective one of the plurality of first trenches, of two adjacent active pillars of the plurality of active pillars.

19. The method for manufacturing the semiconductor structure according to claim 13, further comprising: depositing a first initial dielectric layer in each of the plurality of first trenches; removing the first initial dielectric layer at a bottom portion of each of the plurality of first trenches to remain the first initial dielectric layer arranged on a sidewall of each of the plurality of first trenches to form a first dielectric layer; and doping the bottom portion of each of the plurality of first trenches to form a bit line structure below the bottom portion of each of the plurality of first trenches, wherein the bit line structure extends in the first direction.
